Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Physicist (Ph.D. and M.Sc. in Physics) with ten years of research in computational astrophysics. Taught a graduate thermodynamics course and mentored three graduate researchers who co-authored papers with me. Teaching approach built on intuition, critical thinking, and collaborative problem-solving.

Teaching and Mentoring

Thermodynamics (graduate level) Jun 2018
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Substitute Instructor

Zachary Davis, Amanda O. Harrison, and Hao Zhang, graduate students 2018 – 2022
Department of Physics and Astronomy, Purdue University Mentoring

Education

Universitat de València, *Ph.D. in Physics, Excellent Cum Laude* Jul 2017

Universidad Michoacana de San Nicolás de Hidalgo, *M.Sc. in Physics* Sep 2011

Universidad Autónoma del Estado de México, *B.Sc. in Physics* Dec 2008

Experience

Independent Contractor, *Mercor* Jan 2026 – Present

- Evaluate physics reasoning in AI model outputs, identifying failure modes and providing structured feedback.

Spatial Data Scientist, *TealWaters* May 2025 – Jan 2026

- Mapped wetland probability with a random-forest model (>75% accuracy) and optimized terrain-analysis code.

Independent Researcher & Open-Source Developer, *Self-directed* Jan 2024 – May 2025

- Co-developed Tleco, a Rust/Python toolkit for radiation from relativistic outflows, published in ApJ (2024).

Software Engineer, *Paychex* Apr 2022 – Jan 2024

- Developed and tested production data-pipeline software (Java, Python, SQL).

Postdoctoral Research Associate, *Rochester Institute of Technology* Feb 2021 – Apr 2022

- Led an NSF-sponsored upgrade of a simulation code for black hole binaries; collaborated on a multi-institutional study of binary neutron star mergers (2 publications).

Postdoctoral Research Fellow, *Purdue University* Oct 2018 – Nov 2020

- Co-I in a NASA-funded grant studying radiation from active galaxies; mentored three graduate researchers; published 3 peer-reviewed papers.

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo* Jan – Sep 2018

- Built open-source data analysis tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Outreach

El origen del universo y el sistema solar (The Origin of the Universe and the Solar System) Feb 2026
Biology students, Facultad de Ciencias, Universidad Autónoma del Estado de México Online talk

The Role of Machine Learning in Environmental Science Mar 2024
Seminar on Climate Change Ecology, University of Washington Seattle, WA

Los más rápidos y los más furiosos (The Fastest and the Most Furious) Sep 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco, Mexico Online talk

Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics) Aug 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco, Mexico Online talk

Skills

Physics: Mechanics, electromagnetism, thermodynamics, relativity, statistical physics, astrophysics
Computing: Python, Fortran, C/C++, high-performance/parallel computing
Languages: English (fluent), Spanish (native), Nahuatl (basic)

Selected Publications (9 total, 3 first-author)

Full list available on request.